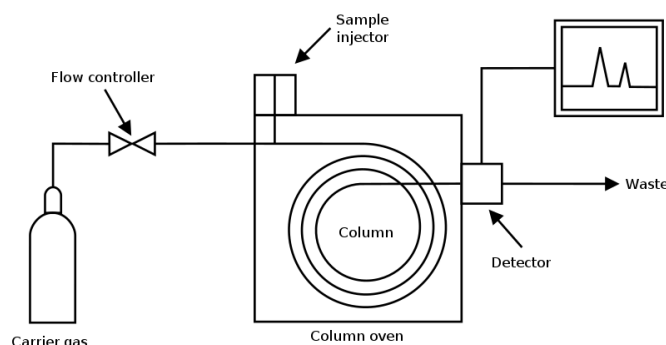


Analytical Tools for Residue Analysis

Gas Chromatography–Mass Spectrometry

Ceramic samples to be analyzed for lipids and other small organic molecules are typically analyzed using gas chromatography-mass spectrometry, commonly abbreviated GC-MS. Gas chromatography is a technique that uses a nonreactive carrier gas (such as helium or argon) to carry molecules extracted from the ceramic sample through a long tube, known as a column. The column is heated to help the molecules move through quickly. A solid layer on the inside of the column slows large or highly charged molecules, allowing smaller and lighter molecules to be separated based on mass and charge. As molecules reach the end of the column, they are introduced into a detector, here a mass spectrometer, for analysis. This method allows all the different molecules (there can be dozens or hundreds!) present in the original sample to be separated and analyzed individually.



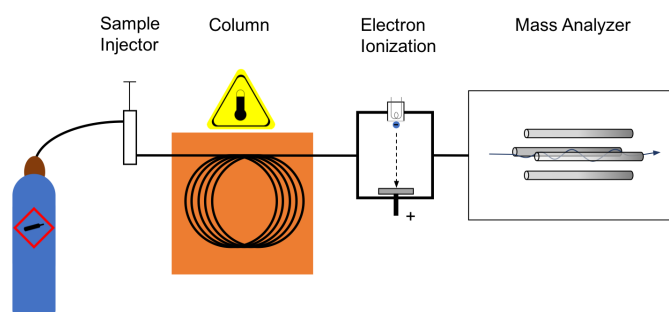
A diagram of a gas chromatograph.

Source: [Wikimedia Commons](https://commons.wikimedia.org/wiki/File:Gas_chromatograph-vector.svg)

(https://commons.wikimedia.org/wiki/File:Gas_chromatograph-vector.svg)

Before a sample can be introduced into a gas chromatograph, the organic compounds (lipids or others) need to be extracted from the ceramic matrix into a liquid or gas state. That is usually done by physically destroying a portion of the ceramic, usually through drilling, to produce a ceramic powder; this vastly increases the surface area of the ceramic. The powder is then washed with organic solvents, such as acetone, capable of dissolving lipids and other nonpolar compounds. That liquid can then be purified and the compounds from the ceramic, concentrated into a sample for GC-MS.

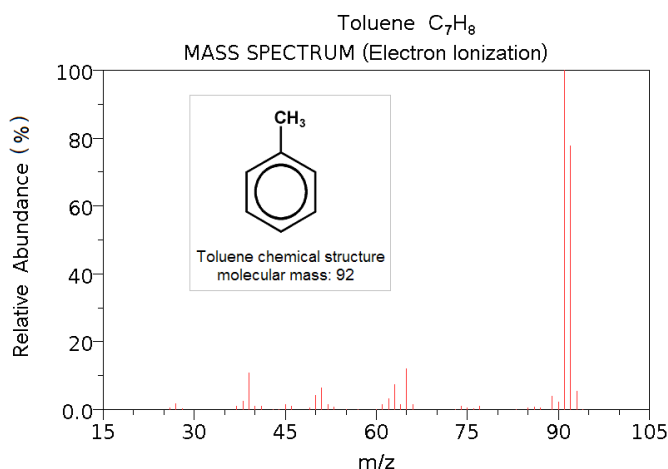
The final step is analysis with a mass spectrometer. A mass spectrometer measures the mass-to-charge ratio of ions (charged particles). It does this by first ionizing the molecule introduced (from the end of the GC column) by hitting it with electrons, causing it to fragment, and those fragments, to become charged. The ionized fragments are then drawn into a set of powerful magnets, which again separate those fragments by size (actually, by mass-to-charge ratio), resulting in the lightest fragments passing through the magnets first, and the heavier fragments following. As they exit the magnets, they hit a detector that counts them. The output is a mass spectrum, a graph that compares the mass-to-charge ratio of the fragments with their count. This graph can be compared with a database containing the mass spectra of known compounds, and a computer can then determine the original chemical structure of the molecule introduced. A mass spectrum is generated for every compound that exits the GC column, allowing a determination of the full range of organic compounds present in the ceramic sample.



A diagram of a GC-MS that uses electron ionization to convert molecules into ions.

Source: [Wikimedia Commons](#)

(https://commons.wikimedia.org/wiki/File:Electron_ionization_GC-MS.png)



NIST Chemistry WebBook (<http://webbook.nist.gov/chemistry>)

A mass spectrum of the aromatic toluene.

"m/z" is the mass-to-charge ratio; the vertical axis has been normalized so that the most frequent ion is set to 100%.

Source: [Wikimedia Commons](#)

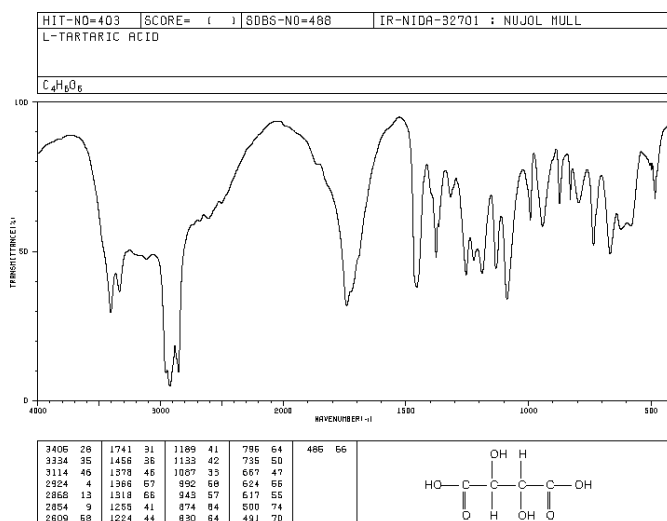
(https://commons.wikimedia.org/wiki/File:Toluene_ei_ms.PNG)

Fourier-Transform Infrared Spectroscopy

Fourier-transform infrared spectroscopy, more commonly known by the abbreviation FTIR, is another method for generating a chemical fingerprint of an unknown compound. It is often less sensitive than GC-MS, as it does not separate the molecules of the sample for analysis, but rather analyzes them all together at once. Watch the five-minute video from Bruker Corporation, one of the main manufacturers of analytical equipment such as FTIR devices, which explains how the technique works. The spectrum that FTIR produces is interpreted through comparison with FTIR spectra from known compounds, similar to the way that mass spectra are interpreted. This analysis process can be automated, with the computer checking peaks against chemical databases and assigning compounds accordingly.



Bruker



An example of an FTIR spectrum: tartaric acid.

Source: SDBSWeb (<https://sdbb.db.aist.go.jp/>) National Institute of
Advanced Industrial Science and Technology